

Practical No. 7

Output:

```
> db.student.createIndex({departement: 1})
< departement_1
> db.student.find({department: "CS"}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'universityDB.student',
    parsedQuery: {
      department: {
        '$eq': 'CS'
      }
    },
  },
  indexFilterSet: false,
  queryHash: 'EB9C5E66',
  planCacheShapeHash: 'EB9C5E66',
  planCacheKey: '6A1E24D9',
  optimizationTimeMillis: 3,
```

```
> db.students.createIndex({ description: "text" })
< description_text
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
  { v: 2, key: { email: 1 }, name: 'email_1', unique: true },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'description_text',
    weights: { description: 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
]
```

```
> db.students.find({ age: { $gt: 25 } }).explain("queryPlanner")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'universityDB.students',
    parsedQuery: {
      age: {
        '$gt': 25
      }
    },
  },
  indexFilterSet: false,
  queryHash: '6ADA4B96',
  planCacheShapeHash: '6ADA4B96',
  planCacheKey: '22477281',
  optimizationTimeMillis: 29,
```

```
> db.students.dropIndex("department_1")
< { nIndexesWas: 2, ok: 1 }
> db.students.getIndexes()
< [ { v: 2, key: { _id: 1 }, name: '_id_' } ]
```

```
> db.students.createIndex({name: 1, age: -1})
< name_1_age_-1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' }
]
```

```
> db.students.createIndex({ email: 1 }, { unique: true })
< email_1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
  { v: 2, key: { email: 1 }, name: 'email_1', unique: true }
]
```

```
> db.students.find({ age: 30 }).explain("allPlansExecution")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'universityDB.students',
    parsedQuery: {
      age: {
        '$eq': 30
      }
    },
  },
  indexFilterSet: false,
  queryHash: '0637AF1A',
  planCacheShapeHash: '0637AF1A',
  planCacheKey: 'D54E87D9',
```

```
> db.students.createIndex({age: 1}, {partialFilterExpression: {age: { $gt: 18}}})
< age_1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
  { v: 2, key: { email: 1 }, name: 'email_1', unique: true },
  {
    v: 2,
    key: { age: 1 },
    name: 'age_1',
    partialFilterExpression: { age: { '$gt': 18 } }
  }
]
```

```

> db.students.createIndex({name: 1}, {sparse: true})
< name_1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
  { v: 2, key: { email: 1 }, name: 'email_1', unique: true },
  { v: 2, key: { name: 1 }, name: 'name_1', sparse: true }
]

```

```

> db.students.reIndex()
< {
  nIndexesWas: 5,
  nIndexes: 5,
  indexes: [
    { v: 2, key: { _id: 1 }, name: '_id_' },
    { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
    { v: 2, unique: true, key: { email: 1 }, name: 'email_1' },
    { v: 2, key: { name: 1 }, name: 'name_1', sparse: true },
    { v: 2, key: { age: 1 }, name: 'age_1', background: true }
  ],
  ok: 1
}

```

```

> db.students.createIndex({age: 1}, {background: true})
< age_1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1, age: -1 }, name: 'name_1_age_-1' },
  { v: 2, key: { email: 1 }, name: 'email_1', unique: true },
  { v: 2, key: { name: 1 }, name: 'name_1', sparse: true },
  { v: 2, key: { age: 1 }, name: 'age_1', background: true }
]

```

Commands:

```
db.students.createIndex({department: 1})
```

```
db.student.find({department: "CS"}).explain("executionStatus")
```

```
db.students.dropIndex("department_1")
```

```
db.students.createIndex({name: 1, age: -1})
```

```
db.students.createIndex({ email: 1 }, { unique: true })
```

```
db.student.getIndexes()
```

```
db.students.explain("executionStats")
```

```
db.students.createIndex({ description: "text" })
```

```
db.students.find({ age: { $gt: 25 } }).explain("queryPlanner")
```

```
db.students.find({ age: 30 }).explain("allPlansExecution")
```

```
db.students.createIndex({age: 1}, {partialFilterExpression: {age: { $gt: 18}}})
```

```
db.students.createIndex({name: 1}, {sparse: true})
```

```
db.students.createIndex({age: 1}, {background: true})
```

```
db.students.reIndex()
```